



Project Title: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

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Business Problem Diagnosis and Understanding

HealthConnect Clinic is an appointment-based healthcare provider experiencing operational inefficiencies driven by high rates of patient missed appointments (no-shows) and unmanaged cancellations. These missed slots lead to unutilized clinical capacity, increased patient wait times, administrative overhead, and lost operational revenue.

The primary objective of this Data Analytics workstream is to evaluate the **4,850** appointment records provided in HealthConnect Appointment Dataset to uncover key patterns behind patients missed appointments and evaluate the structural effectiveness of patient reminders, lead times, clinic wait times, and geographic access.



Data Quality Assessment and Data Cleaning

- An initial inspection of the dataset was conducted in Excel to verify data integrity, understand the data dictionary and variables to have better knowledge of the business activities and operational problems prior to measure creation.
- The data set was checked for completeness, missing values and duplicate. Necessary columns, grouping of distance and time into categories and Dax measures were created to aid better analysis in Power BI.



Exploratory Data Analysis

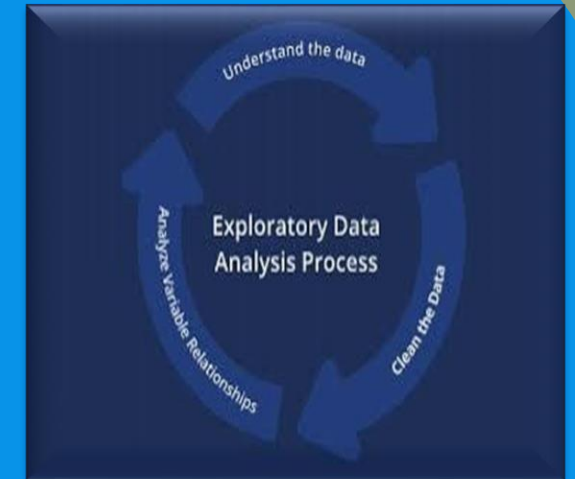
Overall Outcome Distribution: Identified that non-attendance is the primary operational challenge, with 2,339 no-shows slightly exceeding 2,254 attended visits, alongside 257 cancellations.

Spatial Patterns: Over 80% of total appointment volume originates from patients living within 15 km of the clinic (5 - 9.9 km being the largest volume bucket at 1,094 patients). However, propensity analysis revealed that individual no-show risk scales positively with travel distance, rising from 46% for nearby patients to over 54% - 73% for patients residing beyond 20 km.

Communication Channel Variance: Evaluated non-attendance across notification methods, establishing that patients receiving no reminders exhibit the highest no-show propensity (51.28%), whereas automated SMS notifications yield the lowest non-attendance (45.53%).

Demographic Breakdown: Discovered that seniors aged 65+ comprise the largest overall patient volume (437 general consultations) while simultaneously demonstrating the lowest no-show rate (44.90%). In contrast, the 55 - 64 age bracket shows the highest non-attendance propensity (50.97%).

Temporal Invariance: Verified that appointment scheduling windows (Morning at 47.97%, Afternoon at 48.12%, and Evening at 49.39%) show minimal variance in attendance outcomes.



HealthConnect Appointment Analysis

Analyzed by Theresa Onyegiri

Appointment Type

- ☐ Specialist Consultation
- ☐ General Consultation
- ☐ Follow-up
- ☐ Diagnostic Test

Total Appointments

4,850

Total Unique Patients

1,686

No-Show Rate

48.23%

Average Distance (km)

10

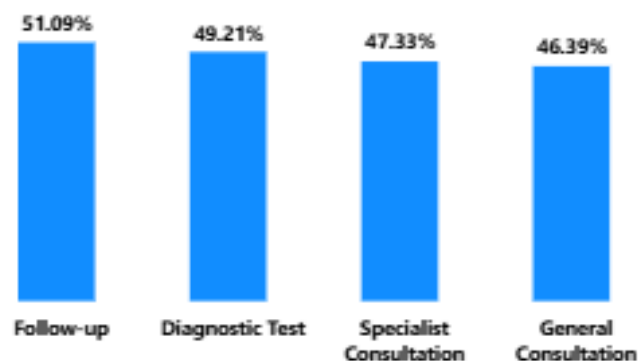
Average Wait Time (min)

24

No-Show Rate by Reminder Channel



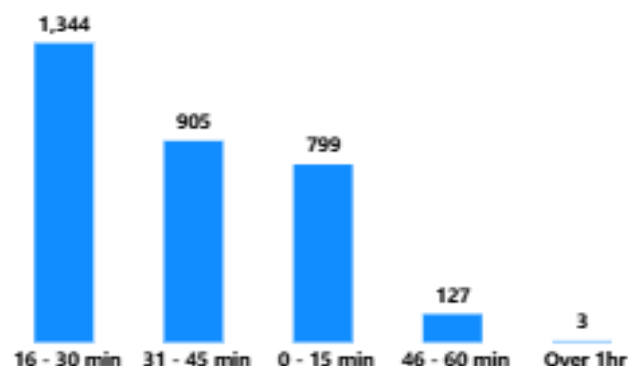
No-Show Rate by Appointment Type



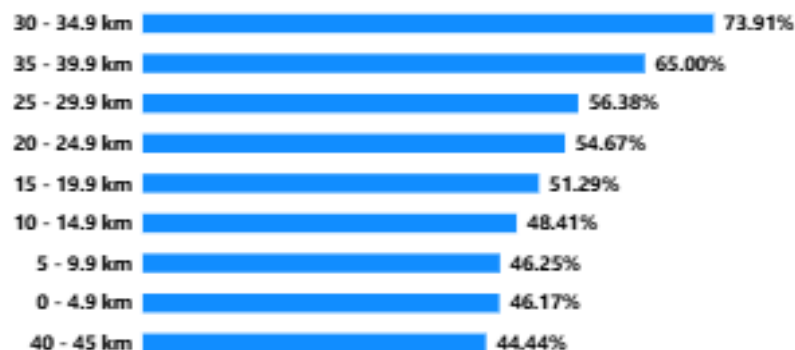
No-Show Rate by Age Group



Total Patients By Waiting Time



No-Show Rate by Distance to Clinic



No-Show Rate by Appointment Time



Cross Track Collaboration and Predictive Model Creation

Collaborated with;
Zamokuhle Maseko
Data Science Intern

Refinement Implemented:

- Designed and integrated an automated logging component. Every inference executed by the model generates a record saved directly into a CSV log.
- Re-engineered the session state and form handling logic to ensure that all user-inputted feature values remain persistent on the dashboard after execution.

The screenshot displays a web application titled "Appointment Prediction" under the heading "APPOINTMENT INTELLIGENCE". The main heading reads "Predict appointment attendance before the appointment." Below this, a sub-header states: "Health Connect uses patient appointment information and behavioural indicators to estimate whether an upcoming appointment is likely to be attended or missed."

The application form is divided into four sections:

- 1 Patient Information:** Includes input fields for "Age" (value: 30) and "Previous No. Shows" (value: 0).
- 2 Appointment Information:** Includes input fields for "Booking Lead Days" (value: 5) and "Waiting Time (minutes)" (value: 10), along with a "Booking Days Category" dropdown menu (value: Medium).
- 3 Distance Information:** Includes input fields for "Distance to Clinic (km)" (value: 30) and a "Distance Category" dropdown menu (value: Far).
- 4 Reminder Information:** Includes a "Reminder Received" dropdown menu (value: Yes).

Below the form sections is a "Logistic Regression Prediction Model" section with a sub-header: "Enter the appointment information above. Health Connect will process the values using the trained appointment attendance model." A "Generate Prediction" button is located to the right of this section.

The prediction result is displayed in a green box with a checkmark icon, stating: "Likely to Attend". Below this, a note reads: "Based on the information provided, the model predicts that this appointment is likely to be attended." At the bottom of the result box, it states: "Prediction confidence: 82.54%".

The footer of the application includes the "Health Connect" logo and the text: "Health Connect Appointment Intelligence Machine Learning Prediction System".

Predictive Model Findings

1. Distance and Historical Behavior Compounding Risk

Distance Threshold: A distance of 15km and above serves as a primary risk boundary across all demographics. When combined with a history of 2 or more previous no-shows, it becomes the strongest compounding predictor of missed appointments.

Geographic Friction: Patients living beyond 20km with booking lead times exceeding 10 days represent the highest overall failure rate, showing that travel burden combined with long waiting periods severely degrades attendance reliability.

2. Reminders act as powerful risk mitigators

High Sensitivity in Older Demographics (55 - 64): Although the 55 - 64 age group exhibits the highest baseline no-show rate (50.97%), they show the highest positive response to intervention. Reminders effectively neutralize the negative impact of long distances even up to 30km, provided the patient has fewer than 2 prior no-shows.

3. Lead time directly impacts patient commitment

The 5-Day Safety Window: Booking lead times under 5 days naturally yield high attendance rates without requiring heavy intervention.

The 10-Day Critical Boundary: Lead times exceeding 5 days begin dropping in reliability, while lead times over 10 days create a critical drop-off point where reminders become strictly mandatory to prevent no-shows.

4. Demographic Profiles Require Tailored Strategies

Younger Patients (18 - 34): Proximity-driven and reminder-sensitive. They display strong natural reliability when living near the clinic (less than 15km), and have high drop-off rates at longer distances (15km and above) if not reminded.

Business Recommendations

- **Prioritize SMS Notifications:** Transition to automated reminder systems to prioritize SMS as the primary touchpoint, as it yields a 5.75% lower no-show rate than sending no reminders. Include 1-click confirmation or cancellation links within messages. Transitioning all non-reminded patients to automated **SMS reminders**, will help in reducing missed appointments as SMS outperforms Email and WhatsApp.
- **Focus on follow-up appointments:** Since follow-ups have the highest no-show rate at **51.09%**, the clinic should introduce a dedicated follow-up reminder process. That involves getting a confirmation from the patients (2 days earlier) that they will attend their appointments. This fosters commitment to attend the appointment.
- **Implement Telehealth for Long-Distance Patients:** Introduce virtual consultations for patients located more than 20 km from the clinic to alleviate transit-related attendance barriers and capture non-attending long-distance volume.
- **Target High-Risk Age Demographics:** Deploy targeted follow-up workflows for patients in the 55 - 64 and 25 - 34 age brackets, such as double confirmation prompts 24 hours before scheduled visits.
- **Strategic Overbooking Calibration:** Apply conservative overbooking models (or dynamic waitlist filling) during peak booking windows to hedge against the clinic's baseline 48% no-show rate.



Thank You